

1. Which of the following pairs correctly classifies the signal $x(t) = \cos(2\pi t) + \sin(\sqrt{2}t)$?
- A) Periodic, Energy signal
 - B) Aperiodic, Power signal
 - C) Periodic, Power signal
 - ☒ D) Aperiodic, Energy signal
2. Which condition must a signal $x(t)$ satisfy to be classified as an energy signal?
- A) $0 < \int_{-\infty}^{\infty} |x(t)| dt < \infty$
 - B) $\int_{-\infty}^{\infty} |x(t)|^2 dt = \infty$
 - ☒ C) $\int_{-\infty}^{\infty} |x(t)|^2 dt < \infty$
 - D) $x(t)$ must be periodic
3. Which of the following is a correct property of the Dirac delta function $\delta(t)$?
- ☒ A) $\int_{-\infty}^{\infty} \delta(t) dt = 0$
 - B) $\delta(t - t_0) = 0$ for all $t = t_0$
 - C) $\int_{-\infty}^{\infty} x(t) \delta(t - t_0) dt = x(t)$
 - D) $\int_{-\infty}^{\infty} x(t) \delta(t - t_0) dt = x(t_0)$
4. If $u(t)$ is the unit step function, then which expression represents the ramp function $r(t)$?
- A) $\int_0^t \delta(\tau) d\tau$
 - ☒ B) $\int_0^t u(\tau) d\tau$
 - C) $\frac{d}{dt} u(t)$
 - D) $\delta(t) * u(t)$
5. Which discrete-time signal is periodic with period N ?
- A) $x[n] = n$
 - B) $x[n] = \cos\left(\frac{2\pi n}{N}\right)$
 - C) $x[n] = e^{-n}$
 - ☒ D) $x[n] = u[n]$
6. Which signal is obtained by sampling a continuous-time sinusoid at a frequency less than the Nyquist rate?
- A) Oversampled
 - B) Time-scaled
 - C) Aliased
 - ☒ D) Windowed

7. Which of the following is a memoryless system?
- A) $y(t) = x(t - 1)$
 - B) $y(t) = x(t) + x(t - 1)$
 - ☒ C) $y(t) = x^2(t)$
 - D) $y(t) = \int_{-\infty}^t x(\tau) d\tau$
8. Which of the following systems is not time-invariant?
- ☒ A) $y(t) = x(t - 3)$
 - B) $y(t) = t \cdot x(t)$
 - C) $y(t) = x(t^2)$
 - D) $y(t) = \frac{d}{dt} x(t)$
9. A system is said to be *BIBO stable* if:
- A) Every bounded input produces an unbounded output
 - B) Unbounded input produces bounded output
 - ☒ C) Every bounded input produces a bounded output
 - D) System impulse response is discontinuous
10. Which system is linear?
- A) $y(t) = x(t)^2$
 - ☒ B) $y(t) = 2x(t) + 3$
 - C) $y(t) = \frac{d}{dt} x(t)$
 - D) $y(t) = \sin(x(t))$
11. Which of the following statements about LTI systems is true?
- A) The output is always a sinusoid
 - B) The system can never be causal
 - ☒ C) The response to any input is the same as the response to an impulse
 - ☒ D) The response to any input can be found using convolution
12. Which of the following is *not* a property of LTI systems?
- ☒ A) Superposition
 - B) Time-invariance
 - C) Convolution
 - ☒ D) Memorylessness

13. The convolution integral of two signals $x(t)$ and $h(t)$ is given by:

A) $\int_{-\infty}^{\infty} x(\tau) + h(t - \tau) d\tau$

☒ B) $\int_0^t x(\tau) h(\tau) d\tau$

C) $\int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$

D) $\int_{-\infty}^t x(t) h(\tau) d\tau$

14. Let $x(t) = u(t)$ and $h(t) = e^{-t}u(t)$. What is $y(t) = x(t) * h(t)$?

A) $y(t) = u(t)$

B) $y(t) = e^{-t}u(t)$

C) $y(t) = 1 - e^{-t}$

☒ D) $y(t) = (1 - e^{-t})u(t)$

15. If $h(t)$ is the impulse response of an LTI system and $x(t) = \delta(t - t_0)$, then the output is:

A) $x(t - t_0)$

B) $h(t - t_0)$

C) $h(t + t_0)$

☒ D) $\delta(t - t_0) * h(t)$

16. An LTI system is causal if and only if:

☒ A) $h(t) = 0$ for $t < 0$

B) $h(t) = 1$ for $t < 0$

C) $h(t) = 0$ for $t > 0$

D) $h(t) = \delta(t)$

17. Which property is guaranteed by the commutative property of convolution?

A) $x(t) * h(t) = x(t - t_0)$

☒ B) $x(t) * h(t) = h(t) * x(t)$

C) $x(t) * h(t) = \delta(t)$

D) $x(t) * h(t) = x(t) + h(t)$

18. If an LTI system is both causal and stable, its impulse response must be:

A) Infinite for all t

B) Zero for $t < 0$ and absolutely integrable

☒ C) Zero for $t > 0$

D) Non-zero only at $t = 0$

19. Which of the following statements is true for an LTI system described by the differential equation $\frac{dy(t)}{dt} + y(t) = x(t)$?

- ☒ A) The system is nonlinear
- ☐ B) The impulse response is $h(t) = e^{-t}u(t)$
- ☐ C) The system is unstable
- ☐ D) The system has memory only for $t < 0$

20. What is the output of an LTI system with impulse response $h(t) = e^{-t}u(t)$, when the input is

- ☒ A) $x(t) = u(t)$
- ☐ B) $y(t) = \delta(t)$
- ☐ C) $y(t) = e^{-t}u(t)$
- ☐ D) $y(t) = 1 - e^{-t}$



POLYTECHNIC

Answer table

1	2	3	4	5	6	7	8	9	10
D	C	A	B	D	D	D	A	C	B
11	12	13	14	15	16	17	18	19	20
C	A	B	D	D	A	B	C	A	D

Cos and Sin Always power signal

1-B

16-A

2-C

17-B

3-B $\rightarrow \omega = 2\pi \times 1000$

18-B

4-B

19-B

5-B

20-B

6-C

7-C

8-C

9-C

10-B

11-B

12-B

13-

14-B

15-B